

Ali Maghami, PhD, PMP

AI, COMPUTER VISION, ROBOTICS & CALIBRATION

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EXECUTIVE PROFILE

Engineering and product leader with 10+ years of experience across industrial automation, robotics, machine vision, and applied AI, including 8+ years delivering computer vision and robotic systems. Experienced leading multidisciplinary teams and complex product lifecycles from customer needs and feasibility through architecture, design, validation, manufacturing and system integration, field deployment, commissioning, and service. Combines credible technical judgement across software, hardware, mechanical, electrical, optical sensing, calibration, quality, and manufacturing with PMP-certified execution discipline. Track record of growing talent, setting technical priorities, managing schedules and budgets, coordinating suppliers and stakeholders, and translating emerging AI technologies into commercially relevant industrial solutions.

SELECTED LEADERSHIP & PRODUCT IMPACT

- Built and led a multidisciplinary Vision, AI, and Robotics team of five engineers; expanded the group from two to five through targeted hiring, onboarding, role definition, mentoring, and technical integration.
- Led architecture and deployment planning for an automated railcar vision platform integrating 40+ cameras, stereo pairs, laser profilers, and mono cameras for real-time localization, measurement, and inspection.
- Inventor on two industrial vision and robotics patents and author/co-author of 10+ publications spanning calibration, robotic guidance, composite inspection, and industrial AI.
- Developed an AI-based calibration solution for cooperative aerospace robots that reduced relative error by more than 80%.

LEADERSHIP & TECHNICAL CAPABILITIES

Engineering Leadership: Team building, talent development, mentoring, technical strategy, roadmaps, resource planning, staffing, budgets, prioritization, risk management, and executive/client reporting

Full Product Lifecycle: Customer discovery, requirements, feasibility, architecture, design, prototyping, verification and validation, manufacturing integration, field trials, commissioning, handover, service, and continuous improvement

Multidisciplinary Systems: Software, AI/ML, robotics, machine vision, optical sensing, mechanical, electrical, controls, quality, suppliers, manufacturing, and service organizations

AI, Vision & Calibration: 2D/3D vision, stereo vision, laser profiling, metrology, point clouds, camera/robot/hand-eye/TCP calibration, deep learning, anomaly detection, VLMs, and synthetic data

Program Execution: PMP, agile product ownership, project planning, milestones, schedules, budgets, technical documentation, customer focus, supplier coordination, and cross-functional delivery

Cloud, Edge & Deployment: Microsoft Azure, AWS, Azure IoT Hub, Blob Storage, Container Registry, Docker, Linux, edge-cloud architecture, containerized deployment, and CI/CD

Software & Engineering Tools: Python, C++, C#, MATLAB, OpenCV, Open3D, PyTorch, TensorFlow, ROS, RoboDK, Git, Blender API, SOLIDWORKS, CATIA, Inventor, OpenUSD, Nvidia Isaac, and AutoCAD

PROFESSIONAL EXPERIENCE

Vision & AI Applications Lead | Hatch Ltd. - Mississauga, ON | Sep 2023 - Present

- Lead a multidisciplinary team of five engineers across machine vision, robotics, AI, and data analytics; helped grow the group from two to five through hiring, onboarding, role definition, mentoring, technical reviews, and delivery support.
- Own end-to-end delivery of industrial vision and AI projects from feasibility and requirements through architecture, development, validation, field trials, commissioning, documentation, and support; manage scope, technical priorities, schedules, risks, resources, client deliverables, and leadership/client reporting.
- Partner with client operations and cross-functional teams spanning automation, controls, electrical, mechanical, software, project management, and construction to convert process needs into executable technical roadmaps, system architectures, and validation plans.
- Architected and supported containerized vision and AI applications across Linux edge systems and Azure-connected environments, including deployment, monitoring, data persistence, and controlled software releases.
- Led architecture and deployment planning for a large-scale automated railcar loading and inspection system integrating 40+ cameras and sensors for real-time 2D/3D localization, detection, measurement, and operational decision support.
- Invented and developed a patented multi-camera 3D vision method for localization and shape estimation of reflective metal surfaces in coating processes, creating differentiated intellectual property for industrial product applications.
- Led development of an AI-powered vision solution for localization, tracking, and speed measurement of in-motion hot metal products, addressing harsh lighting, heat, motion, uptime, and reliability constraints.
- Implemented and maintained GitHub Actions-based CI/CD pipelines to automate software builds, testing, and deployment to Linux edge systems, enabling repeatable and controlled software releases.

Director of Machine Vision Solutions / Co-Founder | TetraGen Robotics Inc. - Winnipeg, MB | Jun 2020 - Oct 2022

- Directed machine vision and robotics product development in an early-stage technology company, defining technical priorities, architecture, development workflows, and user-facing features for customer-oriented perception and calibration tools.
- Led software development for robotic part localization, workspace perception, point-cloud processing, robot guidance, and camera/robot calibration workflows.
- Created an automated 3D perception pipeline for robotic bin picking using synthetic 3D data, deep-learning instance segmentation, target pose estimation, and colored point-cloud analysis.
- Built client-server software that distributed synthetic data generation across multiple networked computers, increasing throughput for AI dataset creation and experimentation.
- Developed GUI-based software for camera calibration, robot TCP/tool calibration, and hand-eye calibration, improving repeatability, usability, and transition from research concepts to deployable tools.
- Supported commercialization through technical demonstrations, validation, documentation, client feedback, and prioritization of practical product capabilities.

Visiting Researcher | National Research Council Canada (NRC) - Montréal, QC | May 2022 - Aug 2022

- Developed a deep-learning method to estimate and compensate relative errors in dual-arm robotic systems, achieving more than an 80% reduction in relative error for aerospace assembly applications.
- Analyzed robotic measurement data, resolved data-quality issues, optimized model architectures, and collaborated within a multidisciplinary research-to-application environment focused on 3D sensing and industrial deployment.

Ph.D. Research Assistant | Intelligent Digital Manufacturing Lab, University of Manitoba - Winnipeg, MB | Jan 2019 - Dec 2023

- Designed a crosshair laser profiler and measurement pipeline for online robot path correction, high-precision guidance, and process adaptation.
- Developed multi-camera vision, target-localization, and online error-correction methods for robotic drilling and advanced aerospace manufacturing.
- Created automated data-generation and multi-input semantic-segmentation workflows for inspection, including multi-light image fusion for small damage and drilled-hole quality detection in CFRP composites.
- Published peer-reviewed research in machine vision, robotic calibration, inspection, deep learning, and advanced manufacturing; translated experimental findings into practical system designs and validation methods.

Engineering Manager | Avin Exon Pars Trading Ltd. - Mashhad, Iran | Nov 2017 - Dec 2018

Medical equipment product development, fabrication, ultrasound service, and customer support

- Oversaw medical equipment product development and fabrication projects from planning through execution and completion; established project goals, schedules, budgets, internal procedures, and resource plans.
- Supervised engineering and technical staff, reviewed design outputs and technical reports, mentored junior team members, and supported professional development and improved engineering performance.
- Coordinated outside suppliers, manufacturers, and contractors to secure reliable components, quality production, and efficient supply chains.
- Directed after-sales support and maintenance programs for medical ultrasound systems, including troubleshooting, upgrades, warranty resolution, customer support, and maintenance planning.
- Managed technical documentation and quality/compliance records and worked directly with the CEO to report project progress, allocate resources, and improve execution.

Design Engineer | Avin Exon Pars Trading Ltd. - Mashhad, Iran | Nov 2017 - Dec 2018

- Designed and optimized mechanical systems and components for medical trolleys, hospital beds, and related healthcare equipment; produced detailed 2D drawings and 3D CAD models with GD&T for manufacturing, assembly, and sourcing.
- Built and tested proof-of-concept prototypes and conducted load, durability, and performance validation to support safety, reliability, and product-quality requirements.
- Worked with cross-disciplinary teams to integrate mechanical designs with electrical and control systems and supported manufacturing, assembly, supplier sourcing, and issue resolution.
- Installed, reinstalled, and warranty-serviced ultrasound scanners used in obstetrics and gynecology; prepared and translated user manuals, service guides, and technical documentation.
- Selected as regional service engineer in Iran for SCANNER and BRAEL medical equipment and completed DRAMINSKI medical ultrasound scanner maintenance and service training in 2017.

EDUCATION

Ph.D., Mechanical Engineering: University of Manitoba, Winnipeg, Canada | 2023

M.Sc., Mechanical Engineering - Applied Mechanics: Ferdowsi University of Mashhad, Iran | 2012

B.Sc., Mechanical Engineering - Solid Mechanics Design: Ferdowsi University of Mashhad, Iran | 2010

CREDENTIALS & PROFESSIONAL DEVELOPMENT

- Project Management Professional (PMP)® - Project Management Institute | 2026
- AWS Certified Solutions Architect – Associate - Amazon Web Services
- Professional Scrum Product Owner I (PSPO I) - Scrum.org | 2026
- NVIDIA Deep Learning Institute: Computer Vision for Industrial Inspection; Fundamentals of Accelerated Data Science; Applications of AI for Anomaly Detection; CUDA Python; CUDA C/C++.

SELECTED PATENTS & PUBLICATIONS

- Maghami, A., Darini, M., Marincic, I., and Zia, A. "Vision method and system for coating processes and systems," WO2026039905A1, 2026.
- Khoshdarregi, M., Alborzi, Y., Maghami, A., Dharia, B. N., and Newman, M. "Method for automated 3D part localization and adjustment of robot end-effectors," WO2023102647A1, 2023.
- Maghami, A., Imbert, A., Côté, G., Monsarrat, B., Birglen, L., and Khoshdarregi, M. "Calibration of multi-robot cooperative systems using deep neural networks," Journal of Intelligent & Robotic Systems, 2023.
- Maghami, A. and Khoshdarregi, M. "Vision-based target localization and online error correction for high-precision robotic drilling," 2024.
- Maghami, A., Salehi, M., and Khoshdarregi, M. "Automated vision-based inspection of drilled CFRP composites using multi-light imaging and deep learning," CIRP Journal of Manufacturing Science and Technology, 2021.

SELECTED RECOGNITION

- University of Manitoba Graduate Fellowship Doctoral Award, 2019-2023.
- Three-time recipient of Edward R. Toporeck Fellowship, 2019-2022.
- International Graduate Student Scholarship, 2019-2021.
- Research Completion Scholarship, 2021.